

Lifesight: Building a unified consumer graph with the support of Google Cloud

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Lifesight leverage
Dataproc and Clou
Bigtable to crunch
scale data, provide
accurate and holistic
customer insight
brands and enterprises
they can make better
business decisions

Lifesight

About Lifesight

Lifesight is a customer intelligence platform that helps brands and enterprises leverage identity resolution and data enrichment to power their customer data strategies. Its industry-leading solution enables the transformation of customer data into actionable insights that help drive business decisions, optimize marketing spend, and improve customer experiences.

Industries: Technology

Google Cloud results

- Reduces query processing time by 70% with BigQuery
- Analyzes petabytes of customer data in minutes for better business insights
- Easily experiment with graph databases on Cloud Bigtable
- Leverages Bigtable for 150,000 writes per second (with 10 nodes)
- Fast turnaround

Organizational Data Search

Location: Singapore

Kubernetes Engine



About Search

Search is a global Google Cloud Premier Partner that helps businesses with end-to-end migration and integration of cloud solutions.



Google Cloud (<https://cloud.google.com/>)

Cloud Bigtable (<https://cloud.google.com/bigtable/>)

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

Cloud Dataproc
(<https://cloud.google.com/dataproc/>)

Kubernetes Engine
(<https://cloud.google.com/kubernetes-engine/>)

BigQuery (<https://cloud.google.com/bigquery/>)

time for ad
hoc
queries
from large
streams of
data

In the world of digital marketing, every piece of data is valuable in driving the success of a campaign. A brand that really knows its customers can reach out to them more effectively, whether it's understanding their purchasing habits or how they feel about the brand.

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*—Amit Nandan, Chief
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To help businesses understand their customers better, Lifesight (<https://www.lifesight.io/>) pieces together a range of customer datasets from various sources, before combining them on its platform to analyze them. "Understanding customers better shouldn't be difficult. Our vision is to become the go-to self-serve platform to help businesses easily leverage big datasets to deliver better customer experiences, optimize marketing spend, and make better data-driven decisions," says Amit Nandan, Chief Strategy Officer at Lifesight. "We aim to give companies a 360-degree view of their customers, and as a data-heavy company, we felt that Google Cloud, and its suite of products, was most suited to our business."

Lifesight's customer intelligence platform processes more than 4.6 billion non-personally identifiable information data points daily. That is why having a reliable platform that can manage the load is paramount. Lifesight needed a technology solution that could support large streams of data and run powerful queries on an ad hoc basis in the most cost-effective way.

Today, Lifesight runs its entire operations on Google Cloud (<https://cloud.google.com/>). BigQuery (/bigquery) and Cloud Bigtable (/bigtable) support its use cases in terms of improving query speed and reducing turnaround time for the ad hoc projects, while

Dataproc (/dataproc) ensures a seamless and secure end-to-end data pipeline.

Migrating to Google Cloud with BigQuery

Lifesight worked with Searce (<https://www.searce.com/>) to manage the migration to Google Cloud. After a four-week proof of concept (POC), the team managed to complete a phase-by-phase migration in just six weeks.

"Our setup was not simple because of the scale and complexity of the infrastructure," says Amit. "The Searce team really made sure they understood our exact requirements and went above and beyond to enable a seamless migration."

The POC replicated two of Lifesight's pipelines with three months worth of data and found that the average query processing time was reduced by 75% with BigQuery. The robust data platform also meant that Lifesight could now analyze petabytes of data within just a few minutes. With that in mind, the Searce team used BigQuery to build a data warehouse and migrated its data processing frameworks. For scalability, Lifesight parks its different application stacks on Google Kubernetes Engine (/kubernetes-engine) (GKE).

Even after the migration was complete, the Searce team continued to propose ways to increase optimization. "The team has never been hesitant to pull in external resources to make sure that they have everything they need to support us. As an

example, even after this migration, they are working with us on ways to improve the optimization of our JanusGraph cluster, which is currently stored on Bigtable," Amit says.

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—Amit Nandan, Chief
Strategy Officer, Lifesight

Ingesting and analyzing terabytes of data at scale

As a company that ingests 40 TB of new data every month, Lifesight relies heavily on Dataproc as a managed service to run all of its Spark jobs, transforming batch and streaming data that come from a wide range of data sources. This includes information such as lat-long data linked to certain consumers or other user-related data like web browsing behavior and purchase habits.

"Having a reliable fully managed cloud service like Dataproc is the most important factor for us, where we are able to ingest and transform data from various sources into a format where it can be consumed by some of our downstream applications," says Amit. As a data company, Lifesight has other pipelines of different streaming jobs running concurrently. It moved its existing Kafka pipelines to Pub/Sub (/pubsub) when it migrated to Google Cloud, which powers one of its applications in near real time.

Recognizing the fast read and write speed of Bigtable, along with its ability to scale up and down seamlessly, Lifesight frequently stores and experiments on it with graph databases from its holistic consumer profiles. As Lifesight continues to grow its JanusGraph cluster, the storage continuously increases as more data is added to

power the graph. "When creating graph databases, you sometimes create millions or billions of vertices and edges to form the relationship of different attributes. When you use JanusGraph and Bigtable as the underlying storage layer, you can both execute fast queries and scale your storage layer independently for the size needed," explains Amit. The team can continue to add nodes as needed without worrying about storage, despite the complexity of the graph implementation.

Data scientists can model their data on the NoSQL database so that all the different attributes are able to be queried against each other, and the team can get more specific results. "With Bigtable, we are able to slice and dice the data that is available to generate very specific datasets quickly, to get a holistic and enriched view of consumers' first party data in almost real time," adds Amit.

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Making holistic customer insights available globally

With the right platforms in place to support growth, Lifesight has its eyes set on the global arena. From APAC to its recent expansion into Europe and the US, it plans to further scale Cloud Bigtable and JanusGraph to support more than 2.3 billion users' data and generate even deeper insights.

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